

Results

Simulation	LRU	FIFO	Random
Sequential	0.0%	0.0%	0.0%
Looping	97.0%	97.0%	97.0%
Random	27.0%	26.0%	27.0%
Mixed	57.0%	42.5%	47.0%
Mixed (size = 4)	9.0%	7.5%	17.0%

- LRU outperforms on the Mixed trace: 57% vs 42.5% (FIFO) and 47% (random)
- All policies tie when locality is very high (looping) or nonexistent (sequential)

Observations

LRU wins on Mixed workloads because it keeps recently re-used blocks in cache

FIFO evicts in arrival order, as it ignores whether a block was recently accessed

Random occasionally outperforms FIFO by avoiding its worst-case eviction patterns

A smaller cache (4 blocks) amplifies differences

From Lecture 23:
Memory stall cycles = Miss Rate $\times$ Miss Penalty;
even small miss rate differences compound into large performance gaps

Conclusion

LRU is the strongest general-purpose policy when temporal locality exists

Policy choice matters most when the working set is larger than the cache

When locality is very high or very low, all three policies perform similarly

Real CPU uses LRU or approximations of it - the results show why

Thank You

for your time
and attention